EXERCISES COURSE

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1. Week 2 (3.1)

Problem 1.1. *Calculate* $\displaystyle\int_0^2 \frac{\mathrm{d}x}{\sqrt{(4-x^2)(9-x^2)}}$.

Solution. Let $x = 2\sin\theta$. The integral becomes

$$\int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{9-4\sin^2\theta}} = \frac{1}{3}\int_0^{\pi/2} \frac{\mathrm{d}\theta}{\sqrt{1-\frac{4}{9}\sin^2\theta}} = \frac{1}{3}F\left(\frac{2}{3}, \frac{\pi}{2}\right).$$

□

Problem 1.2. *Calculate* $\displaystyle\int_0^1 \frac{\mathrm{d}x}{\sqrt{(1+x^2)(1+2x^2)}}$.

Solution. Let $x = \tan\theta$. The integral becomes

$$\begin{aligned}\int_0^{\pi/4} \frac{\sec^2\theta \mathrm{d}\theta}{\sqrt{1+\tan^2\theta}\sqrt{1+2\tan^2\theta}} &= \int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{\cos^2\theta + 2\sin^2\theta}} \\ &= \int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{2-\cos^2\theta}} \\ &= \frac{1}{\sqrt{2}}\int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{1-\frac{1}{2}\cos^2\theta}}.\end{aligned}$$

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Let $\phi = \pi/2 - \theta$. The integral becomes

$$\frac{1}{\sqrt{2}} \int_{\pi/4}^{\pi/2} \frac{\mathrm{d}\phi}{\sqrt{1 - \frac{1}{2}\sin^2\phi}} = \frac{1}{\sqrt{2}} \left\{ F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{2}\right) - F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right) \right\}.$$

□

Problem 1.3. *Calculate* $\displaystyle\int_4^6 \frac{\mathrm{d}x}{\sqrt{(x-1)(x-2)(x-3)}}$.

Solution. Let $u = \sqrt{x-3}$ or $x = 3 + u^2$. The integral becomes

$$2\int_1^{\sqrt{3}} \frac{\mathrm{d}u}{\sqrt{(u^2+2)\,(u^2+1)}}.$$

Let $u = \tan\theta$. The integral becomes

$$\begin{aligned} 2\int_{\pi/4}^{\pi/3} \frac{\mathrm{d}\theta}{\sqrt{2\cos^2\theta + \sin^2\theta}} &= 2\int_{\pi/4}^{\pi/3} \frac{\mathrm{d}\theta}{\sqrt{2 - \sin^2\theta}} = \sqrt{2}\int_{\pi/4}^{\pi/3} \frac{\mathrm{d}\theta}{\sqrt{1 - \frac{1}{2}\sin^2\theta}} \\ &= \sqrt{2}\left\{ F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{3}\right) - F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{4}\right) \right\}. \end{aligned}$$

□

Problem 1.4. *Calculate* $\displaystyle\int \sqrt{\sin x}\,\mathrm{d}x$.

Solution.

$$\begin{aligned} \int \sqrt{\sin x}\,\mathrm{d}x &= \int \sqrt{\cos\left(x - \frac{\pi}{2}\right)}\,\mathrm{d}x \\ &= \int \sqrt{1 - 2\sin^2\left(\frac{x}{2} - \frac{\pi}{4}\right)}\,\mathrm{d}x \\ &= 2\int \sqrt{1 - 2\sin^2\varphi}\,\mathrm{d}\varphi \\ &= 2E(\sqrt{2}, \varphi). \end{aligned}$$

□

Problem 1.5. *Calculate* $\displaystyle\int_0^{2\pi} \sqrt{2 + \sin 2\theta}\,\mathrm{d}\theta$.

Solution.

$$\begin{aligned} \int_0^{2\pi} \sqrt{2 + \sin 2\theta}\,\mathrm{d}\theta &= 2\int_0^{\pi} \sqrt{2 + \sin 2\theta}\,\mathrm{d}\theta \\ &= 2\int_0^{\pi} \sqrt{2 + \cos 2\left(\theta - \frac{\pi}{4}\right)}\,\mathrm{d}\theta \end{aligned}$$

$$
\begin{aligned}
&= 2\int_0^{2\pi} \sqrt{3 - 2\sin^2\left(\theta - \frac{\pi}{4}\right)}\,\mathrm{d}\theta \\
&= 2\sqrt{3}\int_{-\frac{\pi}{4}}^{\frac{3}{4}\pi} \sqrt{1 - \frac{2}{3}\sin^2\theta}\,\mathrm{d}\theta \\
&= 2\sqrt{3}\int_0^{\frac{\pi}{4}} \sqrt{1 - \frac{2}{3}\sin^2\theta}\,\mathrm{d}\theta + 2\sqrt{3}\int_0^{\frac{3}{4}\pi} \sqrt{1 - \frac{2}{3}\sin^2\theta}\,\mathrm{d}\theta \\
&= 2\sqrt{3}\left(E\left(\frac{\sqrt{6}}{3}, \frac{\pi}{4}\right) + E\left(\frac{\sqrt{6}}{3}, \frac{3\pi}{4}\right)\right).
\end{aligned}
$$

□

Problem 1.6. *Calculate the perimeter of an ellipse.*

Solution. The ellipse $x = a\cos\theta$, $4y = b\sin\theta$, $a > b > 0$, has length

$$
\begin{aligned}
L &= 4\int_0^{\pi/2} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = 4\int_0^{\pi/2} \sqrt{a^2\cos^2\theta + b^2\sin^2\theta}\mathrm{d}\theta \\
&= 4\int_0^{\pi/2} \sqrt{a^2 - (a^2 - b^2)\sin^2\theta}\mathrm{d}\theta = 4a\int_0^{\pi/2} \sqrt{1 - e^2\sin^2\theta}\mathrm{d}\theta,
\end{aligned}
$$

where $e^2 = (a^2 - b^2)/a^2 = c^2/a^2$ is the square of the eccentricity of the ellipse. The result can be written as

$$
L = 4aE\left(e, \frac{\pi}{2}\right) = 4aE(e).
$$

For the special case of a circle, $a = b = r$, i.e., $e = 0$, and $E(0) = \pi/2$, and we recover the circumference of a circle: $L = 2\pi r$. □

Remark 1.7. *The term elliptic integral was coined by Count Fagnano (1682-1766) in 1750. He discovered that the arclength of the lemniscate can be expressed in terms of an elliptic integral of the first kind.*

Problem 1.8. *Calculate the arclength of a lemniscate.*

Solution. The lemniscate is the curve:

$$
\left(x^2 + y^2\right)^2 = a^2\left(x^2 - y^2\right)^2,
$$

or in polar form

$$
r^2 = a^2\cos 2\theta.
$$

From $\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2 = \mathrm{d}r^2 + r^2\mathrm{d}\theta^2$,

$$
\begin{aligned}
L = 4\int_{r=0}^{\pi/4} \mathrm{d}s &= 4a\int_0^{\pi/4} \frac{\mathrm{d}\theta}{\sqrt{\cos 2\theta}}, \quad \left(\cos 2\theta = \cos^2 u\right) \\
&= \int_0^{\pi/2} \frac{\mathrm{d}u}{\sqrt{2 - \sin^2 u}} = \frac{1}{\sqrt{2}}\int_0^{\pi/2} \frac{\mathrm{d}u}{\sqrt{1 - \frac{1}{2}\sin^2 u}} = \frac{1}{\sqrt{2}} \cdot F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{2}\right).
\end{aligned}
$$

Thus,

$$L = 4a \cdot \frac{1}{\sqrt{2}} F\left(\frac{1}{\sqrt{2}}, \frac{\pi}{2}\right) = a \cdot 2\sqrt{2}(1.85407) = 5.244102a.$$

□

Remark 1.9. *(Historical note: The rectification of the lemniscate was first done by Fagnano in 1718. The lemniscatus, L. 'decorated by ribbons', was first studied in astronomy in 1680 by Cassini, known as the ovals of Cassini, but his book was published in 1749, many years after his death. The curves were popularized by the Bernoulli brothers in 1694.) Cassini considered more general forms of the lemniscate for whose points the products of the distances to two foci is a constant:*

$$d_1 d_2 = b^2,$$

$$b^4 = r^4 + \frac{a^2}{4} - r^2 a^2 \cos 2\theta.$$

When $b = \frac{a}{\sqrt{2}}$ centered at the origin, we get the ribbon-shaped curve.

Exercise 1.10 (Leave to the reader)**.** *Calculate the finite-amplitude pendulum*

Solution. The equation of motion is:

$$ml\ddot{\theta} = -mg\sin\theta.$$

Let $p = \dot{\theta}$, we have

$$p\frac{\mathrm{d}p}{\mathrm{d}\theta} = -\frac{g}{l}\sin\theta,$$

which yields

$$\frac{p^2}{2} = \frac{g}{l}\cos\theta + C.$$

For the initial condition, $t = 0 : \theta = \theta_0$, $\dot{\theta} = 0$, we have

$$\frac{\mathrm{d}\theta}{\mathrm{d}t} = -\sqrt{\frac{2g}{l}}\sqrt{\cos\theta - \cos\theta_0}.$$

The period, T, is given by

$$\frac{T}{4} = -\sqrt{\frac{l}{2g}}\int_{\theta_0}^{0} \frac{\mathrm{d}\theta}{\sqrt{\cos\theta - \cos\theta_0}}$$

$$\text{or,} \quad T = 4\sqrt{\frac{l}{2g}}\int_{0}^{\theta_0} \frac{\mathrm{d}\theta}{\sqrt{\cos\theta - \cos\theta_0}} = 2\sqrt{\frac{l}{g}}\int_{0}^{\theta_0} \frac{\mathrm{d}\theta}{\sqrt{\sin^2(\theta_0/2) - \sin^2(\theta/2)}}$$

$$= 4\sqrt{\frac{l}{g}}\int_{0}^{\pi/2} \frac{\mathrm{d}u}{\sqrt{1 - k^2\sin^2 u}}, \ \sin\left(\frac{\theta}{2}\right) = \sin\frac{\theta_0}{2} \cdot \sin u, \ k = \sin\left(\frac{\theta_0}{2}\right).$$

Hence

$$T = 4\sqrt{\frac{l}{g}} \cdot F\left(k, \frac{\pi}{2}\right),$$

an elliptic integral. For the special case of smalloscillations, $k = 0$, we get the classical result:

$$T = 2\pi\sqrt{\frac{l}{g}}$$

$\square$

2. Week 3 (3.8)

Problem 2.1. *Calculate the area of regions closed by the following curves.*

(1) *The cycloid:* $x = a(t - \sin t)$, $y = a(1 - \cos t)$ $(0 \le t \le 2\pi,\ a > 0)$ *and* x*-axis.*
(2) *The evolvent of sphere* $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$ $(0 \le t \le 2\pi,\ a > 0)$ *and* $x = a$.
(3) *The rhodonea curve* $r = a\sin 3\theta$.
(4) *The folium of Descartes* $r = \dfrac{3a\sin\theta\cos\theta}{\sin^3\theta + \cos^3\theta}$.

Solution. (1)

$$\begin{aligned} S = \int_0^{2\pi a} y\,\mathrm{d}x &= \int_0^{2\pi} a(1-\cos t)\cdot a(1-\cos t)\,\mathrm{d}\theta \\ &= a^2\int_0^{2\pi}(\cos^2 t - 2\cos t + 1)\,\mathrm{d}\theta \\ &= 3\pi a^2. \end{aligned}$$

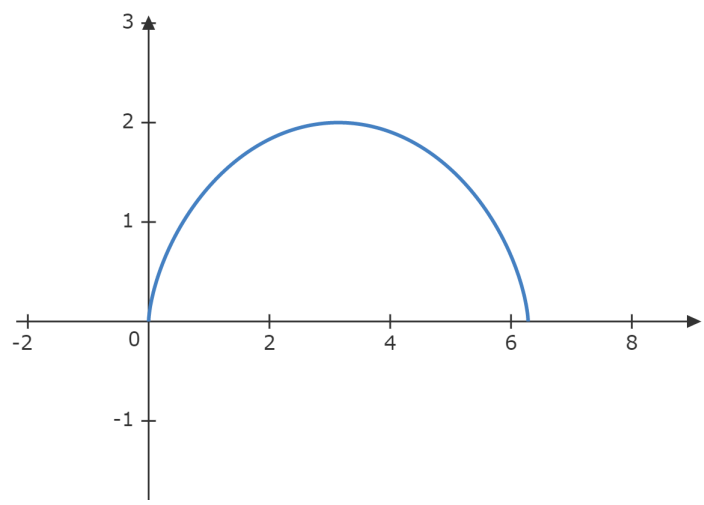

Figure 1. Graph of the cycloid for $a = 1$

(2)

$$\begin{aligned} S = \int (x-a)\,\mathrm{d}y &= a^2\int_0^{2\pi}(\cos t + t\sin t - 1)(\cos t - \cos t + t\sin t)\,\mathrm{d}t \\ &= a^2\int_0^{2\pi}\left(t\sin t\cos t + t^2\sin^2 t - t\sin t\right)\,\mathrm{d}t \\ &= \frac{a^2}{3}(4\pi^3 + 3\pi) \end{aligned}$$

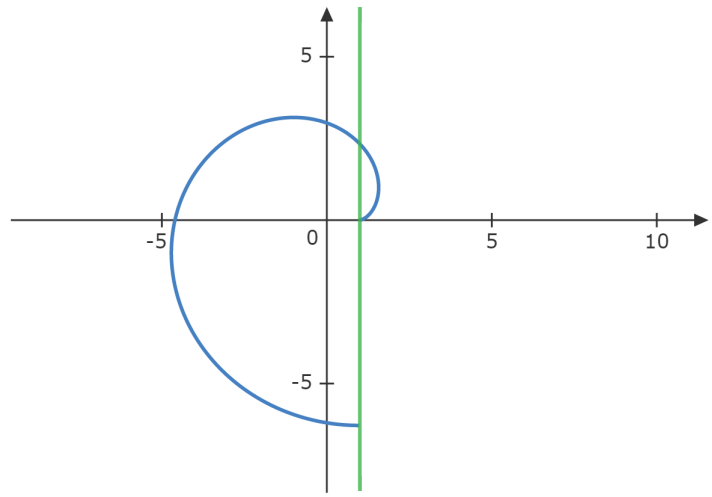

FIGURE 2. Graph of the evolvent for $a = 1$

(3)

$$\begin{aligned} S &= 3 \cdot \frac{1}{2} \int_0^{\frac{\pi}{3}} a^2 \sin^2 3\theta \, \mathrm{d}\theta \\ &= 3 \cdot \frac{a^2}{2} \int_0^{\frac{\pi}{3}} \frac{1 - \cos 6\theta}{2} \, \mathrm{d}\theta \\ &= \frac{\pi}{4} a^2. \end{aligned}$$

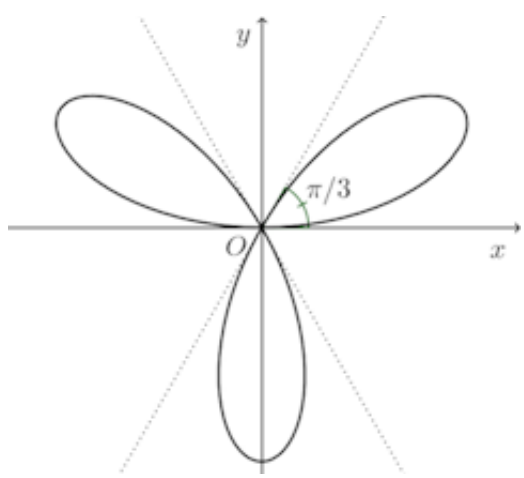

FIGURE 3. Graph of the rhodonea curve for $a = 1$

(4)

$$
\begin{aligned}
S &= \frac{1}{2}\int_0^{\frac{\pi}{2}} \frac{9a^2 \sin^2 \cos^2 \theta}{\left(\sin^3\theta + \cos^3\theta\right)^2}\,\mathrm{d}\theta \\
&= \frac{9a^2}{2}\int_0^{\frac{\pi}{2}} \frac{\tan^2\theta}{\left(\tan^3\theta + 1\right)^2}\,\mathrm{d}\tan\theta \\
&= \frac{3a^2}{2}\int_0^{\frac{\pi}{2}} \frac{1}{\left(\tan^3\theta + 1\right)^2}\,\mathrm{d}\tan\theta \\
&= \frac{3a^2}{2}\frac{-1}{\tan^3\theta + 1}\bigg|_0^{\frac{\pi}{2}} = \frac{3a^2}{2}.
\end{aligned}
$$

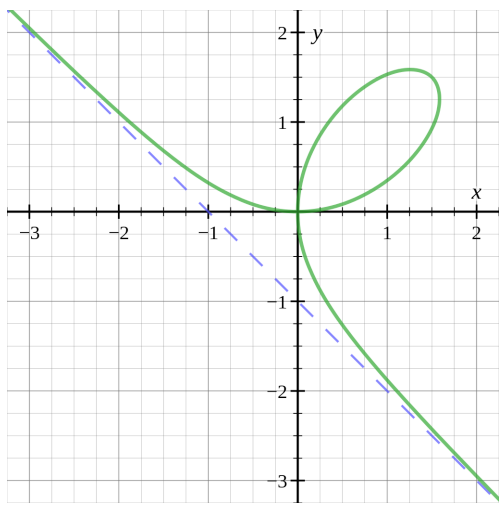

FIGURE 4. Graph of the folium of Descartes for $a = 1$

□

Problem 2.2 (7.43)**.** *Find volumes of the solid of revolution obtained by rotating the region closed by $y = x^2$ $(0 \le x \le h)$, $y = 0$ and $x = h$ around x-axis, y-axis, respectively.*

Solution. Around x-axis:

$$V = \pi\int_0^h y^2\,\mathrm{d}x = \pi\int_0^h x^4\,\mathrm{d}x = \frac{\pi}{5}h^5.$$

Around y-axis: **First way**:

$$V = \pi\int_0^h (h^2 - x^2)\,\mathrm{d}y = \pi\int_0^h (h^2 - x^2)2x\,\mathrm{d}x = \frac{\pi}{2}h^4.$$

Second way:

$$V = 2\pi\int_0^h xy\,\mathrm{d}x = 2\pi\int_0^h x^3\,\mathrm{d}x = \frac{\pi}{2}h^4.$$

□

Problem 2.3 (7.49)**.** *Calculate the arc length of the following curves.*

(2) *Archimedean spiral:* $r = a\theta$ $(0 \le \theta \le 2\pi)$.

(4) *The evolvent of sphere* $x = a(\cos t + t\sin t)$, $y = a(\sin t - t\cos t)$ $(0 \le t \le 2\pi,\ a > 0)$.

Solution. (2)

$$\begin{aligned} L = \int_0^{2\pi} \sqrt{r^2 + r'^2}\,\mathrm{d}\theta &= a\int_0^{2\pi} \sqrt{1+\theta^2}\,\mathrm{d}\theta \\ &= \frac{a}{2}\theta\sqrt{1+\theta^2} + \frac{a}{2}\ln(\theta + \sqrt{1+\theta^2})\Big|_0^{2\pi} \\ &= a\pi\sqrt{1+4\pi^2} + \frac{a}{2}\ln(4\pi + \sqrt{1+4\pi^2}). \end{aligned}$$

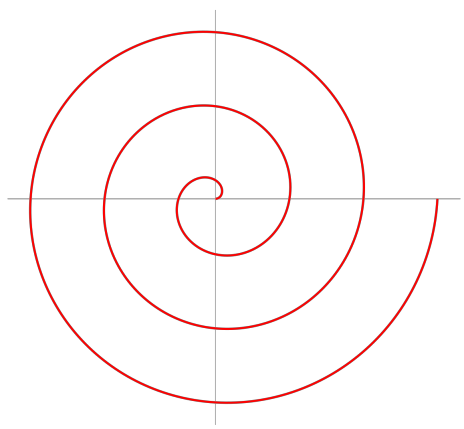

FIGURE 5. Graph of the Archimedean spiral for $a = 1$

(4)

$$\begin{aligned} L = \int_0^{2\pi} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} &= a\int_0^{2\pi} \sqrt{(t\cos t)^2 + (t\sin t)^2}\,\mathrm{d}t \\ &= a\int_0^{2\pi} t\,\mathrm{d}t = 2\pi^2 a. \end{aligned}$$

□

Problem 2.4 (7.51(4))**.** *Calculate the area of the surface of revolution obtained by rotating the curve* $y = \sin x\,(0 \le x \le \pi)$ *around* x*-axis.*

Solution.

$$\begin{aligned} dS &= 2\pi y\sqrt{y'^2 + 1}\,\mathrm{d}x \\ &= 2\pi\sin x\sqrt{1+\cos^2 x}\,\mathrm{d}x. \end{aligned}$$

$$
\begin{aligned}
S &= 2\pi \int_0^{\pi} -\sqrt{1+\cos^2 x}\,\mathrm{d}\cos x \\
&= 4\pi \int_0^1 \sqrt{1+t^2}\,\mathrm{d}t = 2\sqrt{2}\pi + 2\pi \ln(\sqrt{2}+1).
\end{aligned}
$$

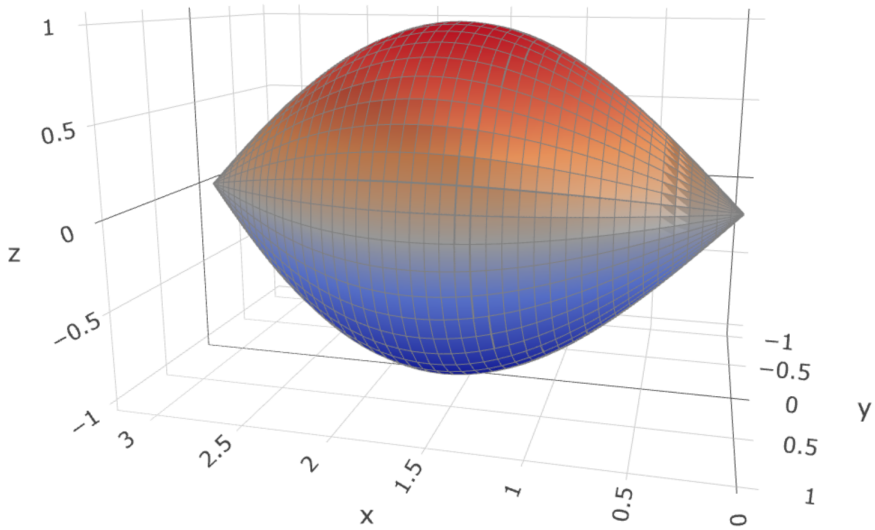

FIGURE 6. $y = \sin x$ rotated around x-axis

□

Problem 2.5. *Find the area of the region determined by* $x^2 + xy + y^2 = 1$.

Solution. **First way:** From the equation, we have

$$
y_{1,2}(x) = -\frac{x}{2} \pm \sqrt{1 - \frac{3}{4}x^2}, \quad -\frac{2}{\sqrt{3}} \le x \le \frac{2}{\sqrt{3}}.
$$

Then

$$
\begin{aligned}
S &= \int_{-2/\sqrt{3}}^{2/\sqrt{3}} [y_1(x) - y_2(x)]\,\mathrm{d}x = 2\int_{-2/\sqrt{3}}^{2/\sqrt{3}} \sqrt{1 - \frac{3}{4}x^2}\,\mathrm{d}x \\
&= \frac{4}{\sqrt{3}} \int_{-\pi/2}^{\pi/2} \mathrm{d}\theta = \frac{2\pi}{\sqrt{3}}.
\end{aligned}
$$

Second way: Let $x = r\cos\theta$, $y = r\sin\theta$, we have

$$
r^2 = \frac{1}{1 + \sin\theta\cos\theta}.
$$

Then

$$
S = \frac{1}{2}\int_0^{2\pi} r^2\,\mathrm{d}\theta = \frac{1}{2}\int_0^{2\pi} \frac{\mathrm{d}\theta}{1+\sin\theta\cos\theta} = \int_0^{2\pi} \frac{\mathrm{d}\theta}{2+\sin\theta} = \frac{2\pi}{\sqrt{3}}.
$$

Third way: Note that

$$1 = x^2 + xy + y^2 = \frac{3}{4}x^2 + \left(y + \frac{x}{2}\right)^2.$$

Let

$$x = \frac{2}{\sqrt{3}}\cos t,\ y = \sin t - \frac{1}{\sqrt{3}}\cos t, \quad 0 \le t \le 2\pi.$$

Since

$$\begin{aligned}
&x(t)y'(t) - y(t)x'(t) \\
&= \frac{2}{\sqrt{3}}\cos t\left(\cos t + \frac{1}{\sqrt{3}}\sin t\right) - \left(\sin t - \frac{1}{\sqrt{3}}\cos t\right)\left(-\frac{2}{\sqrt{3}}\sin t\right) \\
&= \frac{2}{\sqrt{3}},
\end{aligned}$$

we have

$$S = \frac{1}{2}\int_0^{2\pi}(x\,\mathrm{d}y - y\,\mathrm{d}x) = \frac{1}{2}\int_0^{2\pi}\frac{2}{\sqrt{3}}\,\mathrm{d}t = \frac{2\pi}{\sqrt{3}}.$$

□

Problem 2.6. *Suppose that a curve is given by* $y = \int_0^x \sqrt{\sin t}\,\mathrm{d}t$, $0 \le x \le \pi$. *Calculate its arc length.*

Solution. By the formula of arc length calculation, we have

$$\begin{aligned}
L &= \int_0^\pi \sqrt{1 + f'^2(x)}\,\mathrm{d}x = \int_0^\pi \sqrt{1 + \sin x}\,\mathrm{d}x \\
&= \int_0^\pi \left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)\mathrm{d}x \\
&= 2\left(\int_0^{\pi/2}\sin t\,\mathrm{d}t + \int_0^{\pi/2}\cos t\,\mathrm{d}t\right) \\
&= 4\int_0^{\pi/2}\sin t\,\mathrm{d}t = 4.
\end{aligned}$$

□

Problem 2.7. *Use Young's inequality to prove when* $a, b \ge 1$, *there is* $ab \le e^{a-1} + b\ln b$.

Proof. Choosing $f(x) = e^x - 1$, $g(y) = \ln(y+1)$, we have

$$\begin{aligned}
(a-1)(b-1) &\le \int_0^{a-1}(e^x - 1)\,\mathrm{d}x + \int_0^{b-1}\ln(y+1)\,\mathrm{d}y \\
&= e^{a-1} - (a-1) - 1 + b\ln b - (b-1) \\
&= e^{a-1} + b\ln b - a - b + 1,
\end{aligned}$$

i.e.

$$ab \le e^{a-1} + b \ln b.$$

□

Exercise 2.8 (Leave to the reader)**.** *Use Minkowski's inequality to show that*

$$\int_0^\pi |f(x) - \sin x|^2 \, \mathrm{d}x \le \frac{3}{4} \quad \textit{and} \quad \int_0^\pi |f(x) - \cos x|^2 \, \mathrm{d}x \le \frac{3}{4}$$

cannot be simultaneously true, where $f \in R[a, b]$.

Hint:

$$\begin{aligned}
\sqrt{\pi} &= \left(\int_0^\pi |\sin x - \cos x|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} \\
&= \left(\int_0^\pi |(f(x) - \cos x) - (f(x) - \sin x)|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} \\
&\le \left(\int_0^\pi |f(x) - \sin x|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} + \left(\int_0^\pi |f(x) - \cos x|^2 \, \mathrm{d}x \right)^{\frac{1}{2}} \\
&\le \frac{\sqrt{3}}{2} + \frac{\sqrt{3}}{2} = \sqrt{3},
\end{aligned}$$

contradiction. □

Problem 2.9. *Find the minimum of the area enclosed by the parabola* $y^2 = 2x$ *and its chord (passing through the focus).*

Solution. Since the focus of parabola $y^2 = 2x$ is $(\frac{1}{2}, 0)$, we can assume that the equation of a chord is $x = ky + \frac{1}{2}$. Then

$$\begin{aligned}
S &= \int_{y_1}^{y_2} \left[\left(ky + \frac{1}{2} \right) - \frac{y^2}{2} \right] \mathrm{d}y \\
&= \frac{k}{2} (y_2 - y_1) \cdot (y_2 + y_1) + \frac{1}{2} (y_2 - y_1) - \frac{1}{6} (y_2 - y_1) \left(y_2^2 + y_1 y_2 + y_1^2 \right),
\end{aligned}$$

where y_1, y_2 are the intersection points of the parabola and its chord. By $y^2 = 2x$, we have

$$y^2 = 2ky + 1.$$

Hence, the Vieta theorem gives us that

$$\begin{aligned}
y_1 + y_2 &= 2k, \\
y_1 \cdot y_2 &= -1.
\end{aligned}$$

Then

$$
\begin{aligned}
S &= \frac{1}{6}\left(y_2 - y_1\right) \cdot \left[3k\left(y_2 + y_1\right) + 3 - \left(y_2^2 + y_1 y_2 + y_1^2\right)\right] \\
&= \frac{1}{6} \cdot 2 \cdot \sqrt{k^2+1} \cdot \left[3k \cdot 2k + 3 - \left(4k^2 + 1\right)\right] \\
&= \frac{1}{3}\sqrt{k^2+1} \cdot \left(2k^2 + 2\right) \\
&= \frac{2}{3} \cdot \left(k^2 + 1\right)^{\frac{3}{2}}.
\end{aligned}
$$

Hence

$$S_{\min} = \frac{2}{3}.$$

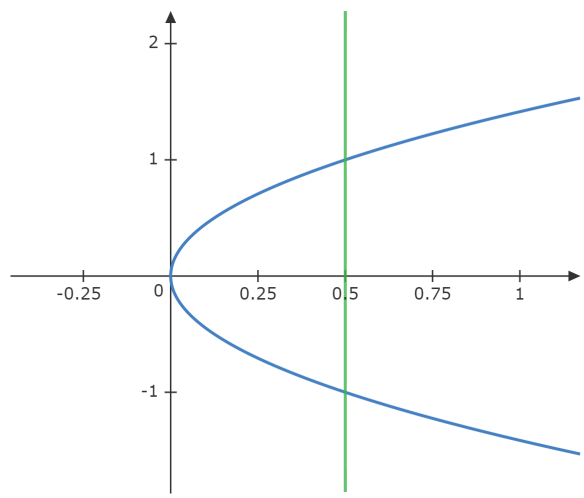

FIGURE 7. Graph of $y^2 = 2x$ and $x = \frac{1}{2}$

□

Problem 2.10 (Wirtinger's inequality)**.** *Suppose that $f : \mathbb{R} \to \mathbb{R}$ is a continuous differentiable function with period 2π, and satisfies $\int_0^{2\pi} f(x)\,\mathrm{d}x = 0$. Prove that*

$$\int_0^{2\pi} f'(x)^2\,\mathrm{d}x \geq \int_0^{2\pi} f(x)^2\,\mathrm{d}x.$$

"=" holds iff $f(x) = a\cos x + b\sin x$.

Proof. Firstly, we show there is a $t_0 \in [0, \pi)$ such that $f(t_0) = f(t_0 + \pi)$. Indeed, if $f(0) = f(\pi)$, we can take $t_0 = 0$. If $f(0) \neq f(\pi)$, let $g(x) = f(x) - f(x + \pi)$, then $g(0)g(\pi) < 0$, hence by the intermediate value theorem, we know that there exists a $t_0 \in [0, \pi)$ such that $g(t_0) = 0$, i.e. $f(t_0) = f(t_0 + \pi)$.

Next, let $c = f(t_0) = f(t_0 + \pi)$. It's easy to calculate that (Leave to the reader!)

$$f'^2 - (f-c)^2 - (f' - (f-c)\cot(t-t_0))^2 = \left((f-c)^2 \cot(t-t_0)\right)'.$$

Note that $(f-c)^2 \cot(t-t_0)$ is continuous at t_0 and $t_0 + \pi$, we have

$$\int_0^{2\pi} f'(t)^2 \, \mathrm{d}t - \int_0^{2\pi} (f(t)-c)^2 \, \mathrm{d}t \geq \left((f-c)^2 \cot(t-t_0)\right)\Big|_0^{2\pi} = 0.$$

So

$$\int_0^{2\pi} f'(t)^2 \, \mathrm{d}t - \int_0^{2\pi} f(t)^2 \, \mathrm{d}t \geq 2\pi c^2 \geq 0.$$

"=" holds iff $c = 0$, and $f'(t) = f(t)\cot(t-t_0)$, then $f(t) = A\sin(t-t_0)$. □

Problem 2.11 (Isoperimetric inequality)**.** *Prove the isoperimetric inequality in $\mathbb{R}^2$: $4\pi A \leq L^2$, "=" holds iff the curve is a circle.*

Proof. Suppose that $\gamma(t) = (x(t), y(t)) : [0, 2\pi] \to \mathbb{R}^2$ is a curve parameterized by arc length. Without loss of generality, we may assume that $\int_0^{2\pi} x(t)\mathrm{d}t = 0$. Then, by Problem 2.10, we have

$$\begin{aligned} L^2 - 4\pi A =& 2\pi \int_0^{2\pi} \left(x'^2 + y'^2\right) \mathrm{d}t - 4\pi \int_0^{2\pi} xy' \mathrm{d}t \\ =& 2\pi \int_0^{2\pi} \left(x'^2 - x^2 + (y' - x)^2\right) \mathrm{d}t \\ \geq& 2\pi \int_0^{2\pi} \left(x'^2 - x^2\right) \mathrm{d}t \geq 0. \end{aligned}$$

"=" holds iff $x(t) = a\cos t + b\sin t, y'(t) = x(t)$, thus it is a circle. □

Exercise 2.12. *Suppose a flat plate of uniform density has the shape contained by $y = x^2$, $y = 1$, and $x = 0$, in the first quadrant. Find the center of mass.*

Hint: We compute the moment around the y-axis:

$$M_y = \int_0^1 x\left(1 - x^2\right) dx = \frac{1}{4},$$

and the total mass

$$M = \int_0^1 \left(1 - x^2\right) dx = \frac{2}{3},$$

and finally

$$\bar{x} = \frac{1}{4} \cdot \frac{3}{2} = \frac{3}{8}.$$

Next we do the same thing to find $\bar{y}$.

$$M_x = \int_0^1 y\sqrt{y} dy = \frac{2}{5},$$

and

$$\bar{y} = \frac{2}{5} \cdot \frac{3}{2} = \frac{3}{5},$$

since the total mass M is the same. $\square$

Remark 2.13. *Since the density is constant, the center of mass depends only on the shape of the plate, not the density, or in other words, this is a purely geometric quantity. In such a case the center of mass is called the centroid.*

Exercise 2.14. *Suppose that a water tank is shaped like a right circular cone with the tip at the bottom, and has height 10 meters and radius 2 meters at the top. If the tank is full, how much work is required to pump all the water out over the top?*

Hint: Here we have a large number of atoms of water that must be lifted different distances to get to the top of the tank. Fortunately, we don't really have to deal with individual atoms–we can consider all the atoms at a given depth together. To approximate the work, we can divide the water in the tank into horizontal sections, approximate the volume of water in a section by a thin disk, and compute the amount of work required to lift each disk to the top of the tank. As usual, we take the limit as the sections get thinner and thinner to get the total work.

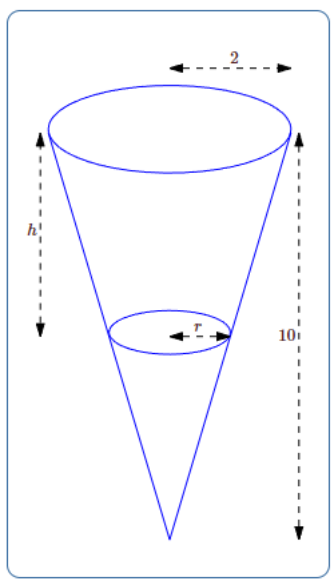

FIGURE 8. A conical water tank

At depth h the circular cross-section through the tank has radius $r = (10 - h)/5$, by similar triangles, and area $\pi(10 - h)^2/25$. A section of the tank at depth h thus has volume approximately $\pi(10 - h)^2/25\Delta h$ and so contains $\sigma\pi(10 - h)^2/25\Delta h$ kilograms of water, where σ is the density of water in kilograms per cubic meter; $\sigma \approx 1000$. The force due to gravity on this much water is $9.8\sigma\pi(10 - h)^2/25\Delta h$, and final1y, this section of

water molst be lifted a distance h, which requires $h9.8\sigma\pi(10-h)^2/25\Delta h$ Newton-meters of work. The total work is therefore

$$W = \frac{9.8\sigma\pi}{25}\int_0^{10} h(10-h)^2 dh = \frac{980000}{3}\pi \approx 1026254 \text{ Newton-meters.}$$

□

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